

Question	Answer	Marks
7(a)	current (through a conductor is directly) proportional to potential difference (across the conductor) or vice versa	M1
	(provided that) temperature (of conductor remains) constant	A1
7(b)(i)	$R = \rho L / A$	C1
	$\rho = (18 \times 7.2 \times 10^{-8}) / 0.94$ $= 1.4 \times 10^{-6} \Omega \text{ m}$	A1
7(b)(ii)	voltmeter reading = 3.1 V	A1
7(b)(iii)	current in the battery: increase	B1
	voltmeter reading: decrease	B1
7(b)(iv)	cross marked on the resistance wire to right of the arrowhead of S, but not touching the right-hand end of the resistance wire	B1
7(c)(i)	$I = Anvq$ $q = 0.93 / [(7.2 \times 10^{-8}) \times (9.0 \times 10^{28}) \times (1.3 \times 10^{-3})]$	C1
	$q = 1.1 \times 10^{-19} \text{ C}$	A1
7(c)(ii)	charge / q (value) is below $1.6 \times 10^{-19} \text{ (C)}$ or charge cannot be below $1.6 \times 10^{-19} \text{ (C)}$ or (the charge carriers / q) should have a charge of $1.6 \times 10^{-19} \text{ (C)}$	B1

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Question	Answer	Marks
2(c)(i)	22: nucleon number or number of neutrons and protons	D1